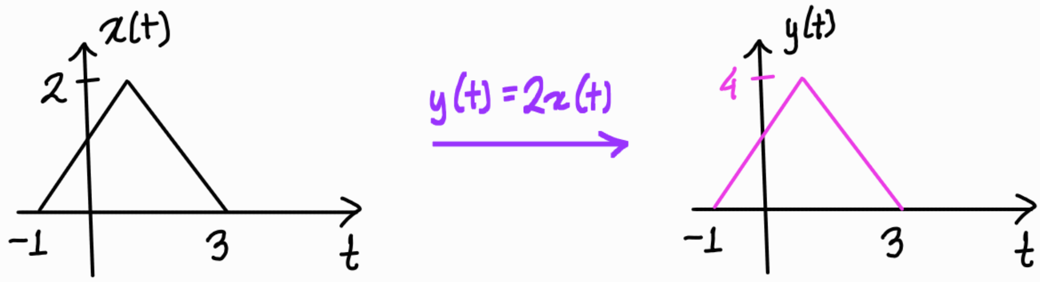
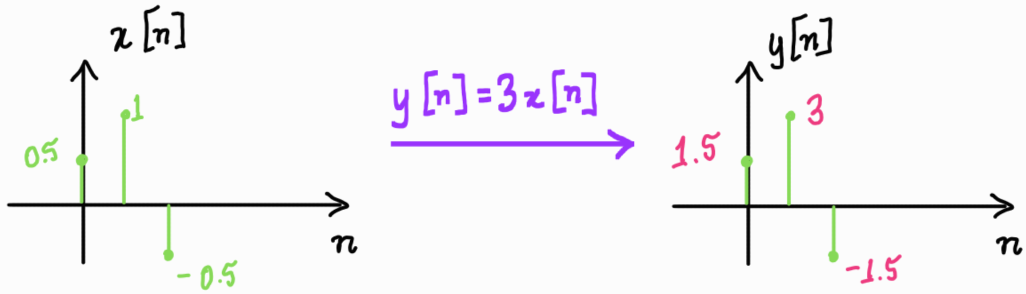
Having learned what a signal is, we will now learn basic signal operations. We will assume all signals to be time dependent.

① Amplitude scaling

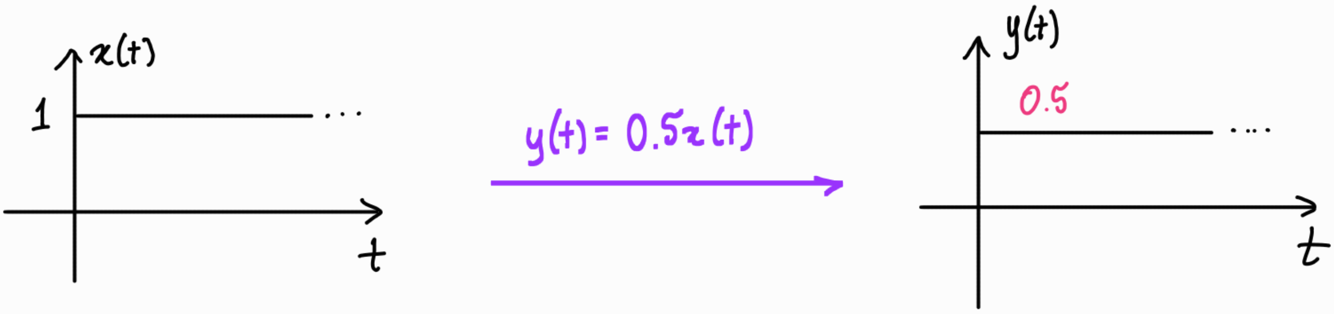
When all points of a signal are multiplied by a constant value, it is called amplitude scaling. General form ⟶ $y(t) = \beta x(t)$ / $y[n] = \beta x[n]$. $|\beta| > 1$ amplifies —



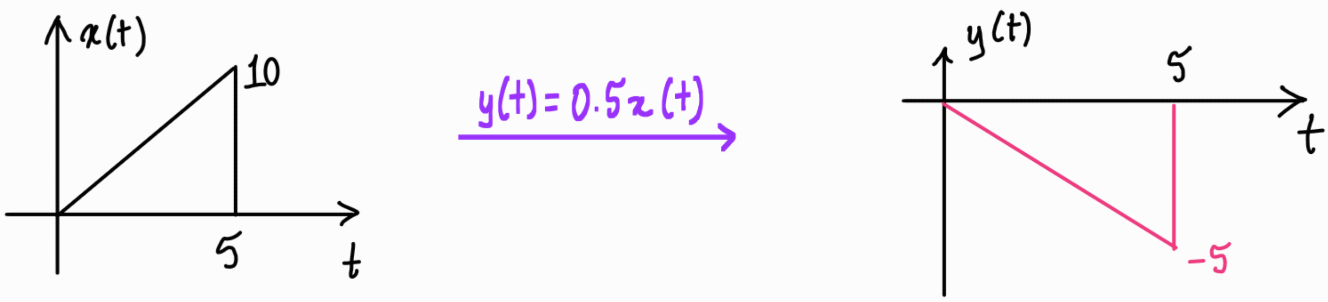


The signal shape remains same. Only the y-axis values get updated.

Amplitude scaling can also compress the signal if the scaling factor, β, is set such that $|\beta| < 1$. For example, if $x(t)$ is given as —

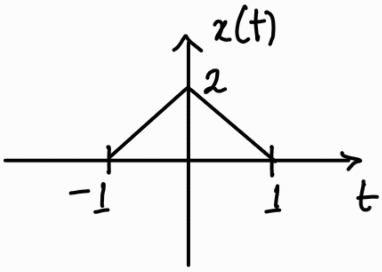


If β is -ve, the signal is reflected over the t-axis / x-axis —

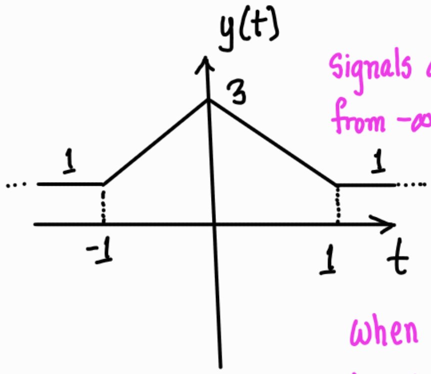


② Amplitude shifting

When a constant value is added to every point of the signal, it is called amplitude shifting. General form $\rightarrow$ $y(t) = x(t) + c \,/\, y[n] = x[n] + c$



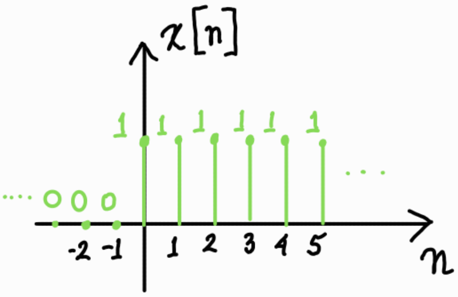
$\xrightarrow{y(t) = x(t) + 1}$



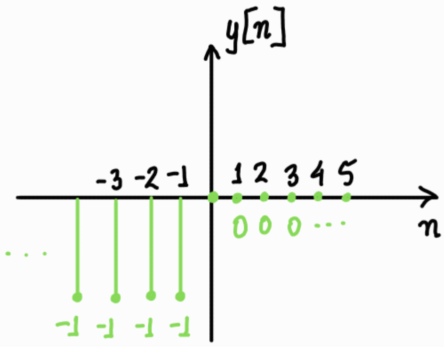
Signals always extend from $-\infty$ to $+\infty$.

When $c > 0$, signal is shifted upward

$x[n] = \{-1, 2, \underset{\uparrow}{-3}, 4, -5\} \xrightarrow{y[n] = x[n] - 2} y[n] = \{\dots, -2, -2, -3, 0, \underset{\uparrow}{-5}, 2, -7, -2, -2, \dots\}$



$\xrightarrow{y[n] = x[n] - 1}$



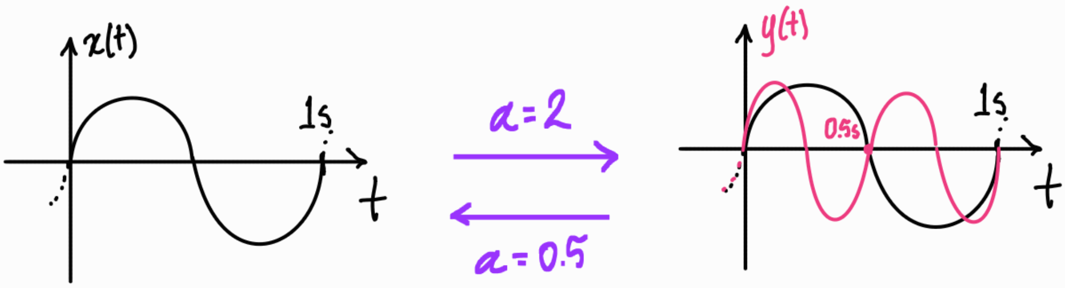
③ Time scaling

Very similar to amplitude scaling, but now the multiplying factor gets placed with t or n. General form $\longrightarrow$ $y(t) = x(\alpha t)$ / $y[n] = x[\alpha n]$. Time scaling is like resampling a signal at a different sampling frequency. This may feel a bit tough and so we will see continuous and discrete time signals separately —

Continuous time signal —

Lets run this with a familiar signal $x(t) = \sin(2\pi t) \longrightarrow$ frequency 1Hz

if $y(t) = x(2t) \longrightarrow y(t) = \sin(2\pi \cdot 2 \cdot t) \longrightarrow$ frequency 2Hz



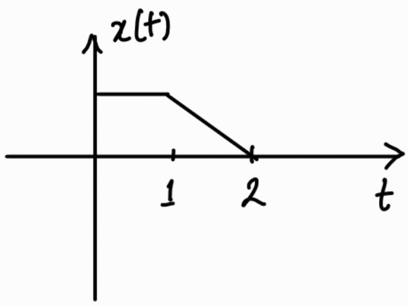
So the signal is effectively getting compressed.

The amplitude remains the same

So if $|\alpha|>1$, the signal gets compressed, multiplied by a factor of $1/\alpha$ in the t-axis

and if $|\alpha|<1$, the signal get expanded, multiplied by a factor of $1/\alpha$ in the t-axis

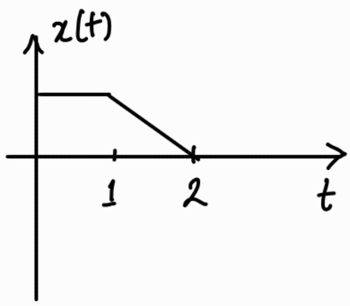
So if you are given an arbitrary signal, what is the easiest way to find the time scaled signal?



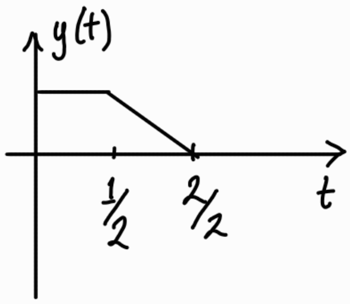
find $y(t) = x(2t)$

Answer —

Redraw the same signal

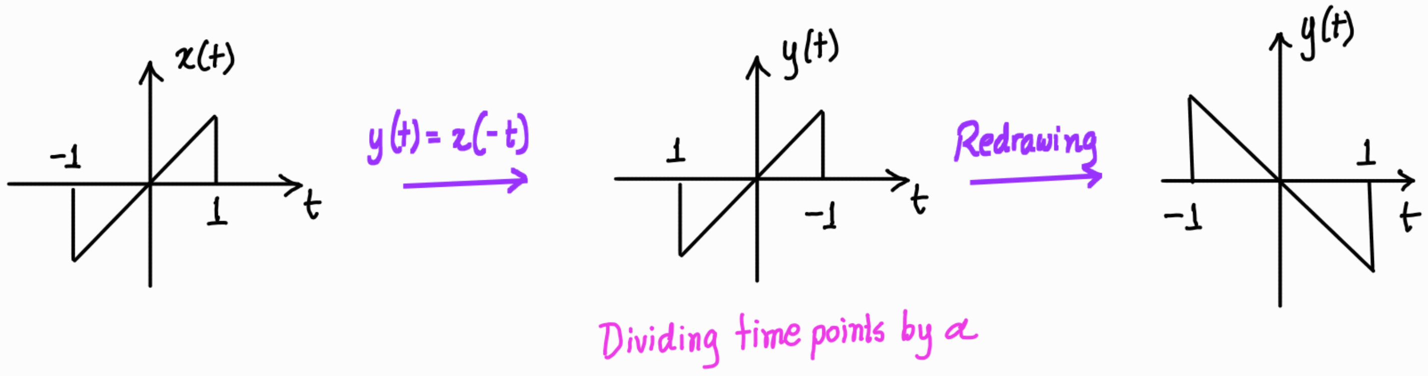


Divide the time points by α



This is the required signal.

When α is -ve, the signal gets reflected over the y-axis. This phenomenon is called time reversal —



You can always do this in one go.

Discrete time signal —

This is a bit different compared to continuous signals. For continuous signals you have infinite points to work with. However for discrete signals you have finite points to work with. Lets work out one problem —

$x[n] = \{3, 4, 7, \underset{\uparrow}{6}, -2, 0, 3, -3\}$, find $y[n] = x[2n]$

Lets do it manually —

for $y[n]$, when $n = -3$, $y[-3] = x[-6] = 0$ ← every unlisted point == 0

when $n = -2$, $y[-2] = x[-4] = 0$

when $n = -1$, $y[-1] = x[-2] = 4$

when $n = 0$, $y[0] = x[0] = 6$

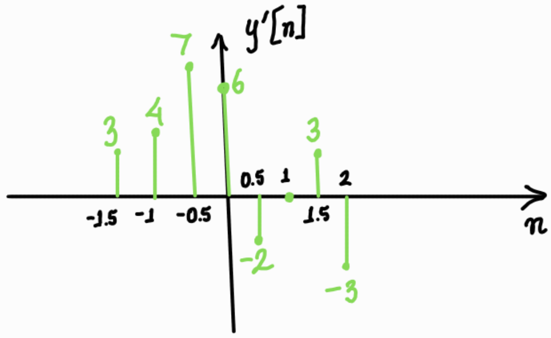
when $n = 1$, $y[1] = x[2] = 0$

when $n = 2$, $y[2] = x[4] = -3$

$\therefore y[n] = \{4, \underset{\uparrow}{6}, 0, -3\}$ ← compression but lossy

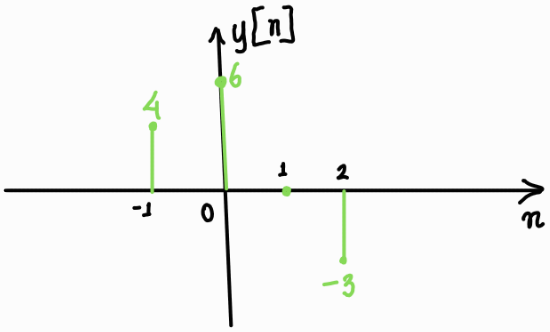
Lets visualize what happens graphically —

$y[n] = x[2n]$



Dividing time points by α. But $n \in \mathbb{Z}$.

So, $\{-1.5, -0.5, 0.5, 1.5\} \notin n$



discarding infeasible points

This is the same signal that we manually derived

What about expansion? We have seen in the continuous case that time scaling is reversible. So lets try it out for discrete case.

Let $x[n] = \{4, \underset{\uparrow}{6}, 0, -3\}$, find $y[n] = x[n/2]$

Lets do it manually —

$y[-4] = x[-2] = 0$

$y[-3] = x[-1.5] = 0$ ⟵ since x doesn't exist at -1.5, the value is 0.

$y[-2] = x[-1] = 4$

$y[-1] = x[-0.5] = 0$

$y[0] = x[0] = 6$

$y[1] = x[0.5] = 0$

$y[2] = x[1] = 0$

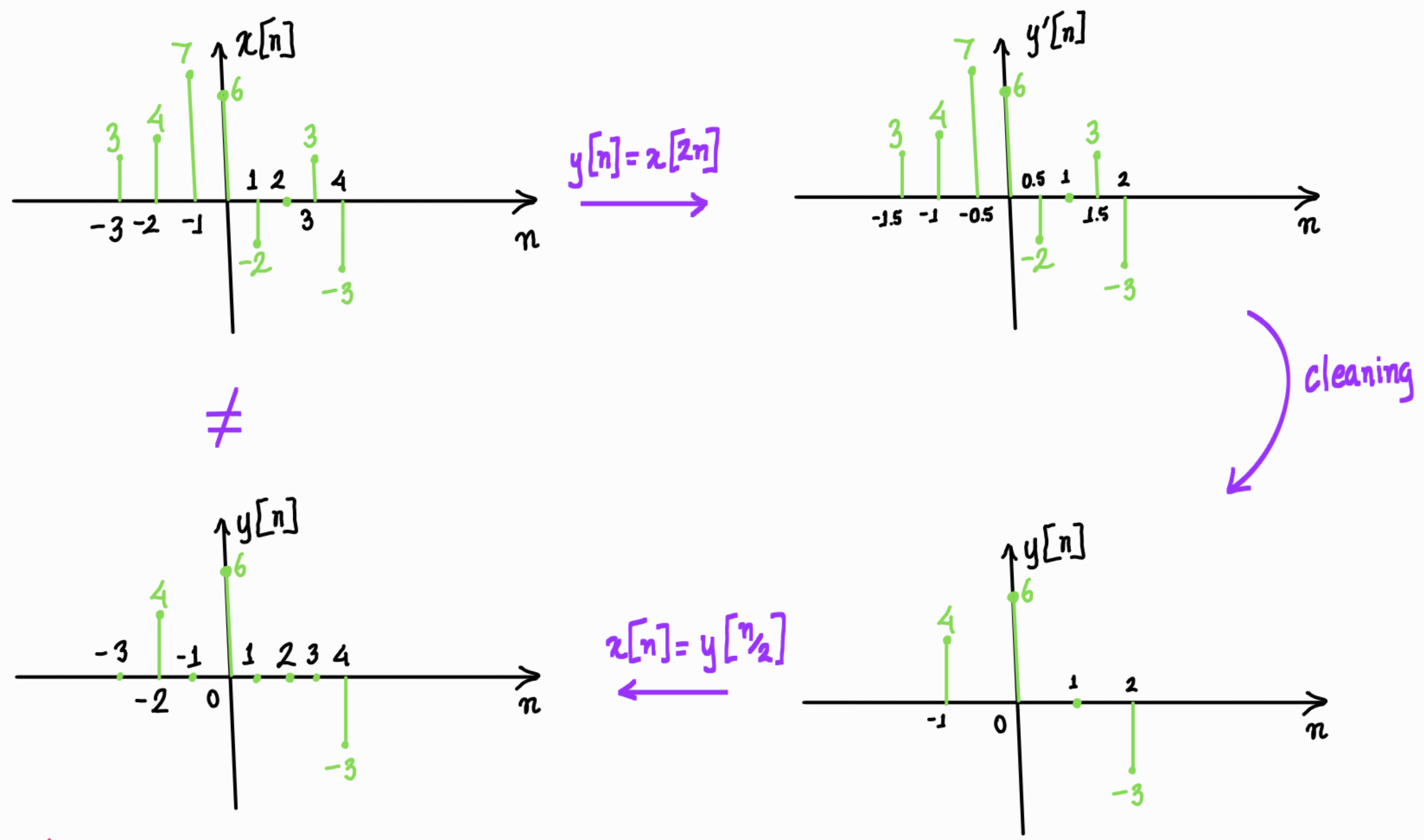
$y[3] = x[1.5] = 0$

$y[4] = x[2] = -3$

$\therefore y[n] = \{4, 0, \underset{\uparrow}{6}, 0, 0, 0, -3\}$

So scaling is not reversible in discrete domain

Lets see the whole thing in action —



You have to fill in the missing n values

4 Time shifting

For time shifting a constant is added to the time variable. General form $\longrightarrow$ $y(t) = x(t+c)$ where $c \in \mathbb{R}$ / $y[n] = x[n+k]$ where $k \in \mathbb{Z}$.

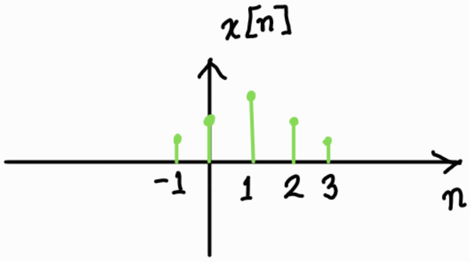
We will learn time shifting using discrete signals, develop a generalized process and go for continuous signals.

Discrete time signals —

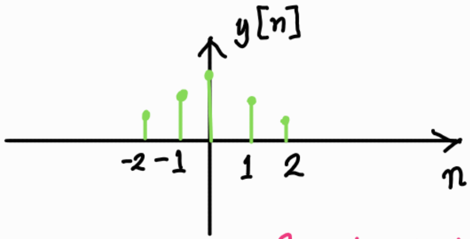
Let $x[n] = \{1, \underset{\uparrow}{2}, 3, 2, 1\}$, then find $y[n] = x[n+1]$

$y[-2] = x[-1] = 1$	$y[0] = x[1] = 3$	$y[2] = x[3] = 1$
$y[-1] = x[0] = 2$	$y[1] = x[2] = 2$	$\therefore y[n] = \{1, 2, \underset{\uparrow}{3}, 2, 1\}$

Graphically —



$y[n] = x[n+1]$



Signal gets left shifted

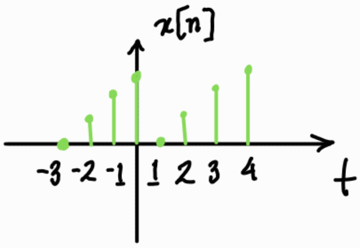
So for +ve time shift $\longrightarrow$ Signal gets left shifted $\longrightarrow$ $n' = n - K$ for $y[n] = x[n+K]$

Similarly it can be shown for -ve time shift signal gets shifted right $\rightarrow$ $n' = n + K$ for $y[n] = x[n-K]$

its a bit counter- intuitive

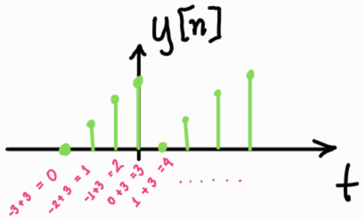
Easiest way to deal with time shift is to consider the standard shifting form to be $x[n-K]$ where K can be +ve or -ve. Then recalculating time points, $n' = n + K$. If $K > 0$, $n' = n + K$ and if $K < 0$, $n' = n - K$, making the process a bit more intuitive.

Lets see an example —



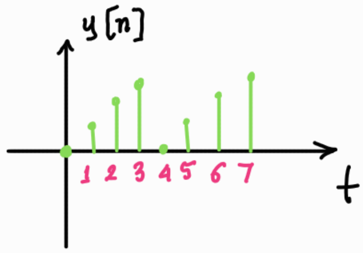
$y[n] = x[n-3]$

comparing to standard form $x[n-K]$, $K = 3$



redrawing

you can always do this in one step



For a discrete signal a fractional K value doesn't make sense. But if K were fractional, you will get a 0 signal.

From here on we will adhere to the standard form — $y[n] = x[n-K]$

$y(t) = x(t-c)$

Continuous time signals —

The concept is the same. But now for $y(t) = x(t - t_0)$, $t_0 \in \mathbb{R}$. So now we can have fractional shift.

for example —